

Business Analytics (Excel + Power BI)– Task 2

Advanced KPI Analysis and Interactive Business Dashboard

Objective

In **Task 1**, you learned how to:

- Clean raw data using Excel
- Create pivot tables and basic charts
- Build a simple Power BI dashboard

In **Task 2**, you will move from **basic reporting** to **business decision-making analytics**.

The objective of this task is to:

- Define meaningful business KPIs
- Perform deeper analysis using Excel formulas
- Use Power BI measures (DAX)
- Build an **interactive dashboard** that answers real business questions

This task introduces you to **how analysts support management decisions**.

Why This Task Matters (Industry Context)

In real companies:

- Managers do not want raw numbers
- They want insights, trends, and comparisons
- Dashboards must be interactive and self-explanatory

A Business Analyst is expected to:

- Translate data into KPIs
- Identify performance gaps
- Enable leadership to take action

This task trains you to think like a **professional Business Analyst**, not just a data operator.

What You Will Build

You will build a **Management-Level Sales Performance Dashboard** that shows:

- Total Revenue and Profit
- Profit Margin by Product and Region
- Monthly and Quarterly Trends
- Top and Bottom Performing Products
- Interactive filters for Region, Category, and Time

This dashboard can be showcased in **interviews and portfolios**.

Dataset Description

You may continue using the dataset from Task 1 or create a new one.

Required Columns

mathematica

Order ID
Order Date
Region
Product
Category
Sales
Profit
Quantity

Concepts You Will Learn

Concept	Explanation
KPIs	Business performance indicators
Advanced Excel Formulas	Deeper data analysis
DAX Measures	Dynamic calculations
Time Intelligence	Trend and growth analysis
Slicers & Filters	Interactive dashboards
Insight Storytelling	Explaining data outcomes

Tools & Technologies Used

- Microsoft Excel
- Power BI Desktop
- Basic understanding of Pivot Tables
- DAX (introductory level)

Prerequisites

Before starting Task 2, ensure:

- Task 1 dashboard is completed
- You understand basic Excel formulas
- You can create charts in Power BI
- Power BI Desktop is installed

Step-by-Step Implementation

Part A: Advanced Analysis Using Excel

Step 1: Create KPI Columns in Excel

Add new calculated columns:

Profit Margin

excel

= Profit / Sales

Sales Category (Using IF)

excel

```
=IF(Sales > 5000, "High", "Medium")
```

Step 2: Create Summary KPI Table

Using Pivot Tables:

- Total Sales
- Total Profit
- Average Profit Margin
- Total Quantity Sold

This table acts as a **business snapshot**.

Step 3: Trend Analysis in Excel

Create Pivot Charts:

- Monthly Sales Trend
- Profit by Region
- Top 10 Products by Sales

These trends help identify:

- Growth patterns
- Seasonal behavior
- Regional performance differences

Part B: Power BI – Advanced Dashboard Creation

Step 4: Load Data into Power BI

- Open Power BI Desktop
- Import Excel file
- Verify data types (Date, Currency, Percentage)

Step 5: Create DAX Measures (Very Important)

Total Sales

DAX

```
Total Sales = SUM(SalesData[Sales])
```

Total Profit

DAX

```
Total Profit = SUM(SalesData[Profit])
```

Profit Margin

DAX

```
Profit Margin = DIVIDE([Total Profit], [Total Sales])
```

Step 6: Time-Based Analysis (Monthly Trend)

DAX

```
Monthly Sales =  
CALCULATE(  
    [Total Sales],  
    VALUES(SalesData[Order Date])  
)
```

Use this measure in:

- Line charts
 - Area charts
-

Step 7: Create Dashboard Visuals

Mandatory visuals:

- KPI Cards (Sales, Profit, Margin)
 - Line Chart (Monthly Sales Trend)
 - Bar Chart (Profit by Region)
 - Table (Top 10 Products)
 - Pie Chart (Category Contribution)
-

Step 8: Add Interactivity

Add slicers for:

- Region
- Category
- Order Date

Ensure:

- All visuals update dynamically
 - Dashboard responds instantly to filters
-

Step 9: Insight Writing (Critical Step)

Write 4–6 insights, for example:

- Region X contributes highest profit but lowest volume
- Product Y has high sales but low profit margin
- Sales peak during specific months

This step separates analysts from dashboard builders.

Key Learning Outcomes

After completing Task 2, you will be able to:

- Define meaningful business KPIs
- Use Excel for analytical calculations
- Write DAX measures in Power BI
- Build interactive dashboards
- Extract and explain business insights

Deliverables

Mandatory:

1. Excel file with calculated columns and pivot tables
2. Power BI .pbix dashboard file
3. 1–2 page PDF containing:
 - KPI definitions
 - Key insights
 - Business recommendations

Optional:

- Screenshot images of dashboard
- Short video walkthrough

Excel – Core Analytics & KPI-Level Skills (Free Resources)

1. Microsoft Excel Official Training (Free)

Complete beginner to intermediate structured lessons from Microsoft.

- Fundamentals, formulas, charts, pivot tables:

<https://support.microsoft.com/excel>

2. Excel for Analytics – Free Course

Covers data analysis, pivot tables, charts, and functions.

- FreeCodeCamp (YouTube, no cost)

<https://www.youtube.com/watch?v=rwbho0CgEAE>

3. Advanced Excel Tutorial (Formulas & Pivot Tables)

- ExcellsFun – Excel Playlist

https://www.youtube.com/playlist?list=PLrRPvpgDmw0ntXRTrxK_wt51YvF9JIWWM

4. Free Excel Formulas Guide (Web)

Great quick reference for essential analytics formulas:

- ExcelJet

<https://exceljet.net/>

Power BI – Free Courses & YouTube Tutorials

1. Microsoft Power BI Guided Learning (Official)

Step-by-step modules covering import, visuals, transformations, and DAX:

<https://learn.microsoft.com/power-bi/guided-learning/>

2. Power BI Full Course for Beginners (YouTube)

Covers data import, visuals, filters, and dashboards:

- Excellent end-to-end tutorial

<https://www.youtube.com/watch?v=AGrl-H87pRU>

3. Power BI Desktop Tutorial for Beginners

Clean walkthrough of UI and report building:

https://www.youtube.com/watch?v=2e9_sS7X4qM

4. DAX Measure Basics (Free YouTube)

Short and focused lesson on creating measures:

<https://www.youtube.com/watch?v=7ppAyi9rQF0>

5. Power BI Community Blog & Tutorials

Official posts on visual design and performance:

<https://community.powerbi.com/>

Free Resources for Specific Analytics Topics

KPI Definitions & Business Insights

- Free article on defining business KPIs
<https://www.klipfolio.com/resources/articles/what-is-a-key-performance-indicator>

Dashboard Best Practices

- Harvard Business Review: What Makes a Good Dashboard
<https://hbr.org/2016/10/the-right-way-to-design-dashboards>

Data Storytelling with Dashboards

Tutorial on how to tell a story with data:

<https://www.tableau.com/learn/articles/data-storytelling>

Templates & Practice Datasets

Excel Practice Files

- Chandoo.org – Free Excel templates
<https://chandoo.org/wp/excel-templates/>

Power BI Sample Data

Official sample datasets for practice:

<https://docs.microsoft.com/power-bi/sample-datasets>

Kaggle Sales Datasets (Free)

Use real datasets for real insights:

<https://www.kaggle.com/datasets>

Essential Tools (Free)

Tool	Purpose
Microsoft Excel (Mobile / Desktop)	Data cleaning, pivot tables, formula analysis
Power BI Desktop (Free Download)	Interactive report building
Google Sheets (Free)	Alternative to Excel
Notion / Google Docs	Documentation and insight writing
Canva / Figma	Optional visuals for presentations

How to Use These Resources for Task 2

For Excel Steps:

- Use Excel courses to master:
 - Pivot tables
 - Advanced formulas (Profit Margin, IF, SUMIFS)
 - Trend charts
 - Conditional formatting

For Power BI Steps:

- Start with official guided learning
- Learn how to:
 - Import & transform data
 - Create measures using DAX
 - Build visuals
 - Add slicers and filters
 - Design an interactive dashboard

For Insights and Explanation:

- Read articles on business KPIs and dashboard best practices
- Practice writing **business recommendations** using Excel and Power BI charts

⚙ Playlist Recommendations (Sequential Learning)

Excel (Beginner → Advanced)

1. FreeCodeCamp Full Course (Pivot Tables + Charts)
<https://www.youtube.com/watch?v=rwbho0CgEAE>
2. ExcellsFun (Formulas-focused)
https://www.youtube.com/playlist?list=PLrPvpvgDmw0ntXRTrxK_wt51YvF9JIWWM

Power BI (End-to-End)

1. Beginner to Advanced Power BI
<https://www.youtube.com/watch?v=AGrl-H87pRU>
2. DAX Measures Introduction
<https://www.youtube.com/watch?v=7ppAyi9rQF0>
3. Visuals & Interactivity
https://www.youtube.com/watch?v=2e9_sS7X4qM